

Hospital Resource Optimization & Data Analytics

Business Problem Bank & Visual Analytics Requirements

This document defines a comprehensive set of data-driven business problems and the corresponding visual analytics requirements for a hospital operations and supply chain analytics project. The intent is to enable independent research and insight generation using structured datasets.

Comprehensive Business Problem Bank

Inventory & Materials Management

- Which medical items contribute to the highest inventory value?
- Which items show low movement but high holding cost?
- Which departments experience frequent stockouts?
- Are reorder levels aligned with actual consumption?
- Which items face high expiry risk?
- What proportion of inventory is non-moving or slow-moving?
- Which categories have the lowest inventory turnover?
- How frequently does emergency procurement occur?
- Is safety stock adequate for critical items?
- How does inventory availability impact treatment delays?

Patient Demand & Capacity Utilization

- What are peak patient admission periods?
- Which departments face demand–capacity mismatch?
- How effectively are beds utilized?
- Which diagnoses lead to longer length of stay?
- Are ICU resources consistently overutilized?
- How predictable is patient inflow over time?
- What seasonal patterns exist in demand?
- Which departments experience congestion?
- What factors drive extended hospital stays?
- How does demand variability stress resources?

Staffing & Workforce Optimization

- Is staff allocation aligned with patient volume?
- Which departments rely heavily on overtime?
- Is workload evenly distributed across staff?
- How does absenteeism affect operations?
- Are night shifts overstaffed or understaffed?
- Which roles face the highest utilization pressure?
- How does staffing adequacy affect patient waiting time?
- Can shift rebalancing improve productivity?

- Are staffing costs proportional to patient throughput?
- Which departments show manpower inefficiencies?

Vendor & Procurement Performance

- Which vendors have frequent delivery delays?
- How does lead time variability affect inventory availability?
- Which vendors contribute to higher procurement costs?
- Are long-term contracts more reliable than spot purchases?
- Which items face vendor dependency risk?
- How do vendor delays trigger emergency procurement?
- Can vendors be segmented by cost and reliability?
- What percentage of orders are delayed?
- Is procurement spend concentrated among few vendors?
- How does order quantity affect delivery reliability?

Financial & Cost Efficiency

- Which departments exceed budgets most frequently?
- What cost categories drive maximum expenditure?
- How much cost is lost due to inventory wastage?
- What is cost per patient by department?
- Are high-cost departments operationally efficient?
- How do emergency purchases impact expenses?
- What is the budget vs actual variance trend?
- Which cost centers lack financial control?
- Can cost overruns be anticipated using trends?
- Which resources show poor cost efficiency?

Integrated & System-Level Problems

- How does patient demand influence inventory consumption?
- How do staffing shortages affect length of stay?
- Do vendor delays lead to service disruptions?
- Which departments face multi-resource bottlenecks?
- How resilient are operations to demand surges?
- Can early warning indicators prevent shortages?
- Which operational decisions have the highest financial impact?

- Are KPIs aligned across departments?
- How does data integration improve decision-making?
- Where do system-level inefficiencies exist?

Visual Analytics Requirements

1. Executive Overview Dashboard

- KPI cards: Total patients, inventory value, staffing cost, operational cost
- Trend line: Patient admissions over time
- Bar chart: Department-wise patient volume

2. Inventory & Supply Chain Analytics

- Bar chart: Inventory value by category
- ABC classification matrix
- Clustered bar: Current stock vs reorder level
- Column chart: Inventory turnover by category
- Table with conditional formatting: Expiry risk items

3. Patient Demand & Capacity Utilization

- Line chart: Admissions trend by department
- Gauge/KPI: Bed utilization rate
- Histogram: Length of stay distribution
- Area/line chart: Peak vs off-peak demand

4. Staffing & Workforce Analytics

- Stacked bar: Staff count by role and department
- Bar chart: Staff utilization rate
- Column chart: Staffing cost by department
- Bar chart: Shift-wise workload distribution

5. Vendor & Procurement Performance

- Bar chart: Vendor on-time delivery rate
- Column chart: Average lead time by vendor
- Bar/Pie chart: Procurement spend by vendor
- Donut chart: Delivery status breakdown

6. Financial Performance & Cost Control

- Bar chart: Department-wise expenditure
- Clustered bar: Budget vs actual spending

- Bar chart: Cost per patient by department
- Line chart: Monthly expenditure trend

7. Integrated & Cross-Functional Analysis

- Combo chart: Patient demand vs inventory consumption
- Combo chart: Staffing levels vs patient volume
- Bar/line chart: Vendor delays vs stockouts
- Heatmap: Operational bottleneck identification

8. Interactivity & Filters

- Date slicer
- Department filter
- Resource type filter
- Vendor filter
- Staff role filter

Expected Outcome: The defined visuals enable proactive monitoring, early issue detection, resource optimization, and evidence-based decision-making across hospital operations.